

SMART INDIA HACKATHON 2026



JALRAKSHA

Problem Statement ID — 26161

Problem Statement Title — Dam Break Inundation Modelling Using Hydrodynamic Modelling of any River

Theme — Disaster Management

PS Category — Software

Team ID — <Team ID from SIH portal>

Team Name (Registered on portal) — <TEAM NAME>



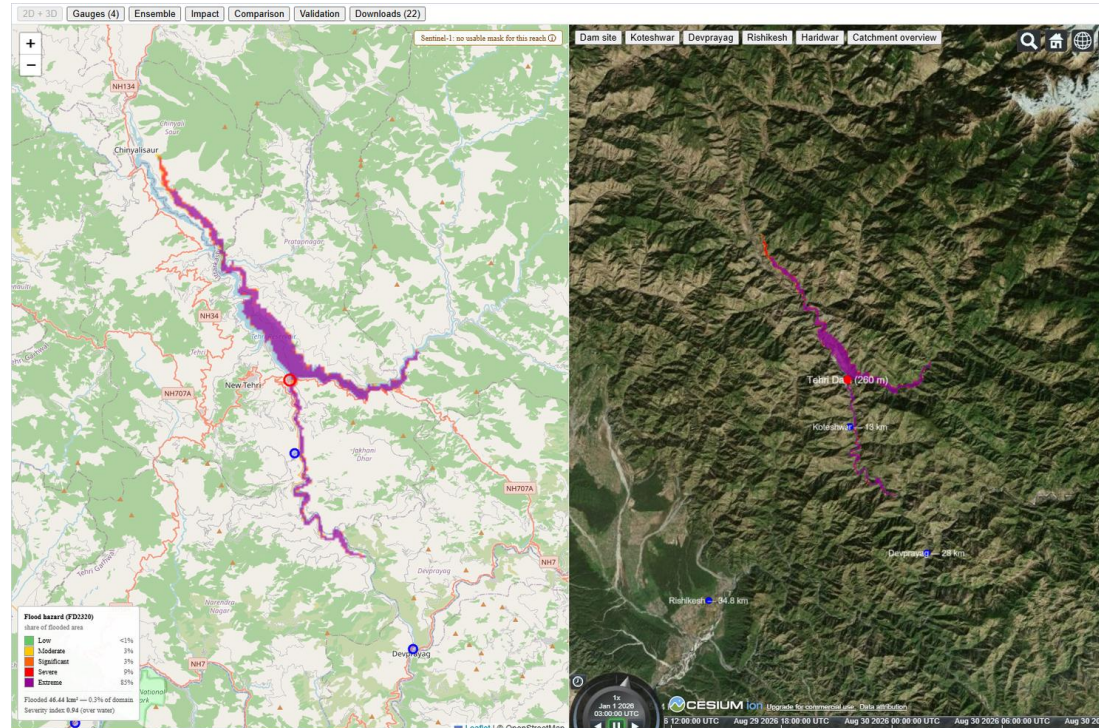
Near-field SPH × far-field 2D shallow water, cross-checked against a real Delft3D FM kernel — arrival-time maps for any Indian dam from nothing but lat/lon, height and storage. Open data only. No ground survey.

JALRAKSHA — dam-break screening, with the model checked

Detailed explanation of the proposed solution

- **Operator enters four things** — lat/lon, dam height, gross storage, breach mode. Nothing else.
- **Data fetches itself:** Copernicus GLO-30 DEM from public AWS COGs, GHSL population, Sentinel-1 SAR via Earth Engine, ESA WorldCover roughness.
- **Breach hydrograph** from a 4-regression Monte-Carlo ensemble — Costa 1985, Froehlich 1995, MacDonald–Langridge 1984, Von Thun–Gillette 1990.
- **Far field:** in-house 2D shallow-water finite volume — HLLC Riemann solver, Audusse hydrostatic reconstruction, MUSCL, wetting and drying.
- **Near field:** Smoothed Particle Hydrodynamics resolves the non-hydrostatic breach jet, where depth-averaging is invalid. One-way handoff, ~600 m over 15 s.
- **Outputs** depth, velocity and arrival-time rasters to a browser dashboard, plus GeoTIFF, .SHP and .KML/KMZ.

The prototype, running



Live dashboard — Tehri (260 m, 3,540 MCM). Left: Leaflet inundation envelope with hazard classes over the Bhagirathi. Right: the same envelope on Cesium 3D over Copernicus terrain, with gauges labelled by downstream distance. 600 × 600 cells at 200 m, 60 km radius, 3 h simulated, EPSG:32644.

How it addresses the problem

- Engineered dam break **and** natural landslide-dam blockage, one pipeline.
- GUI for input and tiled output; **.shp / .kml export** exactly as the PS asks.
- Any DEM, any AOI, any Indian dam — five presets plus custom coordinates.
- Sentinel-1 SAR loop scores simulated extent against **observed** flood extent.
- Runs **offline from cache** after first fetch — demo-day networks assumed hostile.

Innovation and uniqueness of the solution

- **The cross-check is the product.** The same Ritter dam-break runs through our solver and through a real Deltares kernel — *Delft3D FM Suite 2026.01 HM*, *dflowfm-cli* — and both are scored against the exact solution.
- **Uncertainty is shown, not hidden.** The four published breach regressions disagree by a factor of 3–4, so we quote a p05–p95 band, never one number.
- **The tool refuses bad data.** A Sentinel-1 scene scoring 0.486 precision against JRC water is rejected at the 0.50 gate rather than drawn on the map.



TECHNICAL APPROACH



Technologies to be used

SOLVERS

In-house **2D shallow-water finite volume** — HLLC flux · Audusse hydrostatic reconstruction · MUSCL/minmod on $\eta = b + h$ · wetting and drying. **SPH** for the near field. **Delft3D FM** (dflowfm-cli, Suite 2026.01 HM) as the reference kernel.

OPEN DATA

Copernicus **GLO-30** from public AWS COGs · **GHSL P2023A** · **Sentinel-1 GRD** · JRC Global Surface Water · ESA WorldCover · CWC NRLD 2019 / NRSD 2025. None login-gated.

PROOF

Monte-Carlo over four breach regressions with Wahl uncertainty bands. **pytest gates** — analytical solutions, lake-at-rest, mass conservation and the Delft3D FM cross-check — all runnable from the dashboard, all currently passing.

TECHNOLOGY STACK



Python



NumPy



Numba



GDAL



FastAPI



SQLite



React



Vite



Leaflet



Cesium



Earth Engine



pytest

Methodology and process for implementation

1 INPUT

lat/lon · height · storage
breach mode · ensemble size

2 AUTO-INGEST

GLO-30 COGs · GHSL · WorldCover
Sentinel-1 · CWC registers

3 BREACH HYDROGRAPH

4 regressions, Monte-Carlo
p05–p95 117,061–353,612 m³/s

4a NEAR FIELD — SPH

Lagrangian, mesh-free. The violent,
non-hydrostatic breach jet, where
depth-averaging is invalid. ~600 m, 15 s.

4b FAR FIELD — 2D SWE

Eulerian finite volume, HLLC + Audusse
+ MUSCL. 600 × 600 cells at 200 m,
60 km radius, 3 h simulated.

One-way handoff only (SWE → SPH at breach time). No two-way feedback is claimed.

5 CROSS-CHECK — the step most entries skip

Ritter vs the exact solution: **0.0317 m** (JalRaksha) against **0.0349 m** (Delft3D FM) · lake-at-rest 5.98e-14 m/s · mass conservation 0.000000%

6 IMPACT

GHSL population by warning lead time
1,760 at risk — 193 under 15 min

7 OUTPUT

dashboard · arrival-time map
GeoTIFF · .SHP · .KML/KMZ

8 GEE VALIDATION LOOP

Sentinel-1 observed extent scored against the simulation. Scenes below 0.50 precision against JRC water are refused, not drawn.



Analysis of the feasibility of the idea

Not a proposal — a **built system**. These gates run against the live build, and are the same checks that block a merge in CI:

Blocking gate	Measured	Gate
Lake at rest — spurious velocity	5.98e-14 m/s	< 1e-8
Mass conservation — volume drift	0.000000 %	< 0.1 %
Ritter dam-break — RMSE vs exact	0.0317 m	< 0.10 m

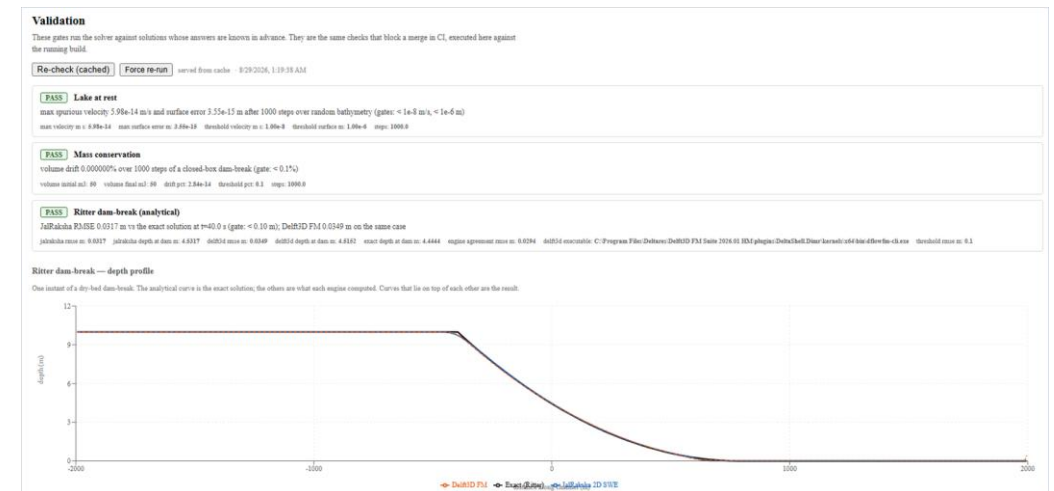
Both engines against the exact Ritter solution, same grid, same instant:

Engine	RMSE vs exact	Depth at dam
JalRaksha 2D SWE	0.0317 m	4.532 m
Delft3D FM (dflowfm-cli)	0.0349 m	4.515 m

Potential challenges and risks

- **Wet-dry front instability** — negative depths and spurious velocities. This fails visibly in a live demo.
- **30 m GLO-30 is a surface model** with no reservoir bathymetry; it flattens water and puts canopy in the channel. 1–2 cells span a Himalayan gorge.
- **Breach regressions disagree by 3–4 ×** — the dominant uncertainty term, and a documented state of the art rather than a defect in this implementation.
- **SPH does not scale** — uniform resolution is an acknowledged limitation (SPHERIC Grand Challenge 3); it cannot span a basin.
- **Debris, not clear water** — Chamoli was ~80% rock by volume, and clear-water physics understates such an event.
- **Manning's n is uncertain**, and inundation extent is sensitive to it.

The gates, run from the dashboard



All three PASS. $h_0 = 10$ m, $t = 40$ s, $\Delta x = 10$ m, frictionless flat bed; exact $4h_0/9 = 4.444$ m, and the two engines agree to 0.0294 m RMSE.

Strategies for overcoming these challenges

- **Positivity-preserving scheme** — Audusse reconstruction under a CFL $\leq 1/2$ bound; lake-at-rest and mass conservation gate every merge, and both pass now.
- **Lead with arrival time and extent**, not point depth. Every depth figure here is labelled indicative; the DEM limit is stated, not buried.
- **Quote the band, never the number** — Monte-Carlo over all four regressions returns p05 / median / p95 for outflow, depth and arrival time.
- **Confine SPH to a near-field box** (~600 m, 15 s) and hand off one-way; the far field is left to the depth-averaged solver, where it is valid.
- **Positioned as a Tier-1 screening layer** under CWC CDSO_GUD_DS_05: it triages which dams warrant a full Tier-2/3 study, and does not replace one.



IMPACT AND BENEFITS



1,180,102

people in the modelled domain
(GHSL P2023A, epoch 2020)

1,760

reach 0.1 m depth or more — the
population actually at risk

193

have under 15 min of warning at a
30 min lead-time assumption

50.2 min

for the wave to reach Koteswar,
13.0 km downstream

46.44 km²

inundated envelope, severity
index 0.94 over water

One real run, not an illustration: Tehri Dam (260 m, 3,540 MCM), central breach, 600 × 600 cells at 200 m over a 60 km radius, 3 h simulated. Peak breach outflow 229,952 m³/s (p05–p95 117,061–353,612); breach formation 29.3 min; maximum depth anywhere 99.9 m (75.1–120.6 m). Depth on a 30 m DEM is indicative — lead with arrival time.

Potential impact on the target audience

- **NDMA / SDMA / DDMA** — depth, extent and arrival time are exactly what an evacuation order needs; .kml drops into the GIS they already run.
- **CWC / NDSA** — a Tier-1 screening layer under CDSO_GUD_DS_05, to triage which of India's large dams warrant a full expert study first.
- **NTRO / HADR planners** — scenarios for dams and landslide lakes that cannot be ground-surveyed, including reaches with no site access.
- **Army / BRO / NDRF** — pre-positioning, and deliberate-breach planning of the kind executed at Phuktal in 2015.
- **Downstream communities** — the people counted in the 1,760 above, who at Chamoli in 2021 received no warning at all.

Benefits of the solution (social, economic, environmental)

SOCIAL

Arrival time is lead time, and lead time is evacuation. The exposed population is split by how many minutes each group actually has — 193 of them under 15. Works for terrain that cannot be surveyed or reached at all.

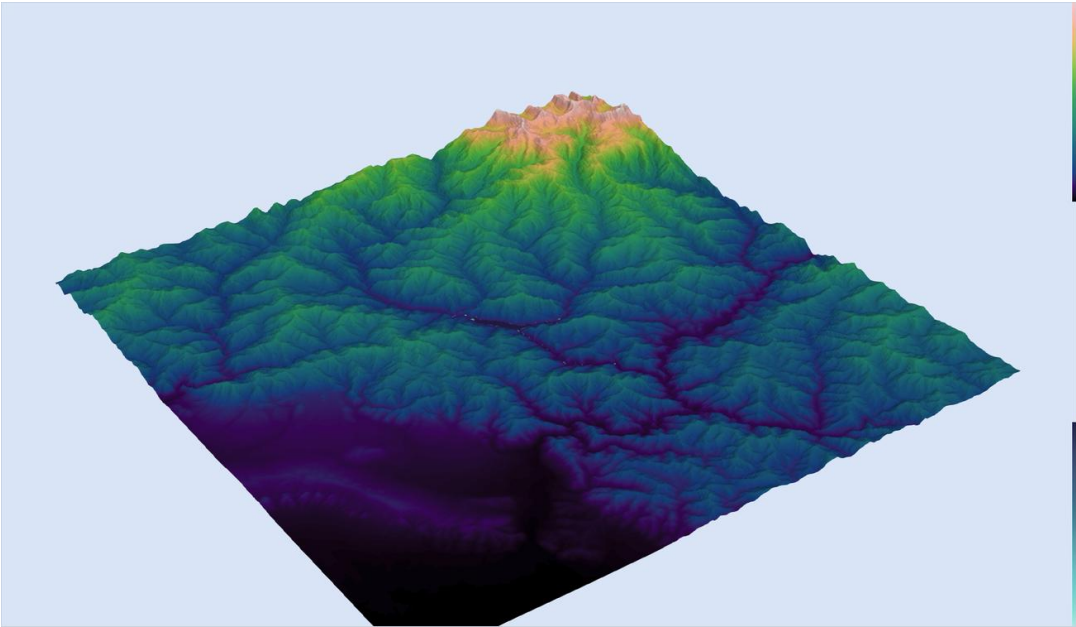
ECONOMIC

Zero licence cost, and automation replaces weeks of specialist setup per site. .shp / .kml at the 1:50,000 and 1:10,000 scales CWC's Emergency Action Plan guidelines specify — usable, not merely viewable.

ENVIRONMENTAL

Identifies cropland and ecosystems in the flood path, and supports scheduled controlled release over emergency release. Open and reproducible: anyone can re-run the cross-check and get these numbers.

Real terrain, real reach



Bhagirathi valley below Tehri on Copernicus GLO-30, rendered in ParaView with the computed depth draped over it. Rishikesh and Haridwar report **no arrival** — Rishikesh sits 18 m above the nearest channel. Reported as a null, not dropped.



RESEARCH AND REFERENCES



Details / Links of the reference and research work

Problem domain — Indian events and regulation

- CWC / CDSO, **Guidelines for Mapping Flood Risks Associated with Dams**, CDSO_GUD_DS_05 v1.0, 2018 — the Tier 1–3 framework this tool targets
- CWC, **Emergency Action Plan guidelines** — index maps 1:50,000, detailed maps 1:10,000
- Shugar et al. 2021, **Science** 373:300–306, doi 10.1126/science.abh4455 — Chamoli; travel times matched to under 5%
- Rana et al. 2023, **Natural Hazards**, doi 10.1007/s11069-023-05972-5 — an independent reconstruction of the same event
- Sinha 2009, **Current Science** 97:429–433 — the 2008 Kosi avulsion
- Costa & Schuster 1988, **GSA Bulletin** — formation and failure of natural dams
- CWC **NRLD 2019** and **NRSD 2025** national dam registers

Numerical methods

- Toro 2001, **Shock-Capturing Methods for Free-Surface Shallow Flows**, Wiley — HLLC wave speeds, two-rarefaction estimate
- Audusse et al. 2004, **SIAM J. Sci. Comput.** 25:2050 — hydrostatic reconstruction, the well-balanced property this solver rests on
- Froehlich 2008, **J. Hydraul. Eng.** 134(12) — embankment breach regressions
- Xu & Zhang 2009, **J. Geotech. Geoenviron. Eng.** 135(12)
- Wahl 2004, **J. Hydraul. Eng.** 130(5) — uncertainty of breach parameters
- Maranzoni & Tomirotti 2023, **Water** 15(17):3130, doi 10.3390/w15173130 — recommends exactly this coupled 2D–3D decomposition
- Vacondio et al. 2020, **Comp. Particle Mech.** 8:575 — SPH Grand Challenge 3
- Ramachandran et al. 2021, **ACM TOMS** 47(4), doi 10.1145/3460773 — PySPH

Open data and software — free, none login-gated

- Copernicus **DEM GLO-30** — 30 m, public AWS Cloud-Optimized GeoTIFFs
- **Sentinel-1 GRD** and **JRC Global Surface Water** via Google Earth Engine
- **GHSL P2023A** gridded population, epoch 2020 (JRC)
- **ESA WorldCover** — land cover driving Manning's n
- **Google Open Buildings** — CC BY 4.0
- **Delft3D FM**, Deltares — Suite 2026.01 HM, dflowfm-cli
- **PySPH** — BSD licence, developed at IIT Bombay
- *Deliberately excluded:* FABDEM (CC BY-NC-SA), MERIT (CC BY-NC / ODbL) and OSM (ODbL share-alike) are kept out of redistributed outputs on licence grounds.

Reproduce every number on this deck: `python scripts/validate_against_delft3d.py --case ritter | jalraksha run --dam tehri | dashboard at localhost:3000`